

- (iii) R, as this has the largest fractional uncertainty. B1
- 2 (a) 15 m s^{-1} A1
- (b) constant gradient (straight line graph) A1
- (c) (i) 1.55 s A1
- (ii) distance = area under the graph from 0 to 1.55 s
 $= \frac{1}{2} (15)(1.55)$ M1
 $= 11.6 \text{ m}$
 $= 12 \text{ m}$ A0
- (iii) distance = $\frac{1}{2} (25)(4.1 - 1.55) - 11.6$ C1
 $= 31.875 - 11.6$
 $= 20 \text{ m}$ A1
- (d) displacement is the straight line / minimum distance between the start and finish points B1
in that direction.
distance is the actual total path travelled by the ball. B1
- (e) Smooth curve with decreasing gradient until zero at terminal velocity B1
gradient of the curve at x-intercept (0 m s^{-1}) = gradient of the straight line
and the curve crosses the x-axis before 1.55 s . B1
- 3 (a) (i) force is the rate of change of momentum B1